

VASTMARKUM, Sweden

WMO: 022600

Lat: 63.1243N

Lon: 18.2518E

Elev: 469

StdP: 14.45

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.1	(c) -4.2	(d) -13.4	(e) 2.7	(f) -6.2	(g) -7.9	(h) 3.7	(i) -2.0	(j) 16.2	(k) 32.7	(l) 14.5	(m) 30.3	(n) 1.8	(o) 310	(p) 0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 20.3	(c) 76.8	(d) 64.5	(e) 73.3	(f) 61.9	(g) 70.2	(h) 60.3	(i) 66.2	(j) 73.5	(k) 63.9	(l) 70.6	(m) 61.9	(n) 67.8	(o) 5.7	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.4	89.0	69.1	61.3	82.3	66.5	59.3	76.8	64.6	31.1	73.4	29.4	70.4	27.9	67.7	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 14.5	(b) 12.6	(c) 11.2	(e) -14.0	(f) 80.8	(g) 5.7	(h) 4.1	(i) -18.1	(j) 83.8	(k) -21.5	(l) 86.2	(m) -24.7	(n) 88.5	(o) -28.8	(p) 91.5		
(a) 14.5	(b) 12.6	(c) 11.2	(e) -13.5	(f) 68.5	(g) 4.7	(h) 2.6	(i) -16.9	(j) 70.4	(k) -19.7	(l) 72.0	(m) -22.3	(n) 73.5	(o) -25.7	(p) 75.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.6	19.3	19.5	26.2	35.3	45.0	52.6	58.5	55.6	47.6	37.3	29.8	23.1		
	DBStd	15.53	10.33	11.06	8.97	5.95	6.32	5.14	4.93	4.98	5.48	6.28	7.70	10.26		
	HDD50	5149	951	855	737	440	181	31	1	10	108	394	607	834		
	HDD65	10014	1416	1275	1202	890	618	373	211	293	522	858	1057	1299		
	CDD50	619	0	0	0	0	27	108	264	184	36	0	0	0		
	CDD65	11	0	0	0	0	0	0	9	2	0	0	0	0		
	CDH74	244	0	0	0	0	5	38	164	37	0	0	0	0		
	CDH80	26	0	0	0	0	0	4	21	1	0	0	0	0		
	WSAvg	4.4	4.0	4.2	4.9	4.6	4.9	5.2	4.3	3.9	4.1	4.4	4.1	4.2		
	PrecAvg	29.3	2.4	1.5	1.7	1.7	1.5	2.2	3.5	3.6	2.8	3.1	3.0	2.4		
(15) Precipitation	PrecMax	48.5	4.9	3.5	4.0	4.8	3.0	5.5	8.8	6.5	7.8	8.9	10.2	4.6		
	PrecMin	19.2	0.3	0.3	0.2	0.2	0.0	0.4	0.4	0.7	0.7	0.6	0.9	0.2		
	PrecStd	6.1	1.3	0.8	0.9	1.1	0.7	1.1	1.9	1.9	1.9	2.0	2.2	1.1		
	PrecStd	6.1	1.3	0.8	0.9	1.1	0.7	1.1	1.9	1.9	1.9	2.0	2.2	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.5	45.0	51.8	63.2	74.2	79.4	82.8	78.3	67.3	55.8	49.0	43.6		
		MCWB	39.3	38.7	42.9	50.3	57.9	60.0	66.4	65.2	58.6	50.8	46.0	40.4		
	2%	DB	39.3	41.7	46.3	56.7	68.3	73.1	78.5	74.1	63.5	52.7	44.7	40.2		
		MCWB	37.0	37.5	39.3	46.8	56.0	58.5	64.8	63.7	56.4	49.3	43.4	37.6		
	5%	DB	36.4	38.2	43.0	51.3	63.8	69.2	74.7	70.9	60.9	50.2	42.0	37.9		
		MCWB	34.2	35.1	37.3	42.8	53.2	56.9	63.6	62.1	55.2	47.7	40.8	36.1		
	10%	DB	33.3	34.6	39.6	47.4	59.5	65.9	71.4	67.5	58.2	48.0	40.2	35.9		
		MCWB	32.3	32.7	35.0	40.1	50.1	56.0	61.8	60.4	54.1	45.9	39.0	34.5		
	0.4%	WB	39.5	39.6	43.2	51.1	60.3	63.1	69.3	68.7	61.0	52.5	47.0	41.0		
		MCDB	42.7	43.5	50.5	61.3	71.0	71.8	76.4	74.4	64.9	54.2	48.5	43.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.1	37.9	40.2	47.1	57.2	60.9	67.2	65.7	58.3	50.2	43.6	38.5		
		MCDB	39.3	41.4	45.6	56.1	66.5	69.2	75.0	72.0	61.4	51.8	44.6	39.8		
	5%	WB	34.7	35.6	37.6	43.5	53.7	59.0	65.1	63.3	56.5	48.3	41.2	36.4		
		MCDB	36.1	38.0	42.2	50.2	62.5	66.9	72.7	68.3	59.4	49.9	41.6	37.6		
	10%	WB	32.6	33.1	35.3	40.8	51.0	56.7	63.0	61.3	54.8	46.1	39.3	34.6		
		MCDB	33.6	34.6	39.2	46.3	58.4	63.3	69.9	65.7	57.3	47.7	40.1	35.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.7	15.8	17.6	18.0	20.2	20.5	20.3	19.5	17.3	13.6	9.8	12.4		
		MCDBR	14.5	16.2	20.0	26.6	29.1	28.2	26.6	25.7	21.9	17.5	11.5	13.2		
		MCWBR	12.7	13.3	13.8	16.6	17.2	15.4	14.9	15.8	15.2	13.9	10.2	12.4		
		MCDBR	13.6	14.6	19.3	24.9	27.0	23.4	23.6	21.8	18.4	15.8	10.9	12.4		
	5% WB	MCWBR	12.3	12.2	13.9	16.2	16.6	14.2	14.3	14.3	13.9	13.5	10.1	11.5		
		MCWBR	12.3	12.2	13.9	16.2	16.6	14.2	14.3	14.3	13.9	13.5	10.1	11.5		
(40) Clear-Sky Solar Irradiance	taub		0.223	0.283	0.307	0.344	0.344	0.341	0.358	0.349	0.318	0.304	0.371	0.345		
	taud		2.071	2.232	2.328	2.385	2.463	2.501	2.471	2.490	2.562	2.451	2.764	2.660		
	Ebn at Noon		132	208	253	266	277	280	271	263	252	205	124	77		
	Edh at Noon		10	19	25	30	30	30	30	27	21	16	9	6		
	RadAvg		51	236	706	1200	1588	1812	1660	1260	764	315	81	26		
(45) All-Sky Solar Radiation	RadStd		7	44	80	110	167	116	164	103	82	55	17	2		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-220	N/A	N/A	N/A

Nomenclature: See separate page